

error. methodology

$$15 = 1111$$

$$225 = 11110001$$

KMAP
Sum of Product
POS

$$128 \times 16 \times 32$$

$$11010001$$

$$168421$$

Verilog
System Verilog
VHDL

* Task:

7 Segment LED, iski bases py KMAP design krna hy.
(7 KMAP), with 4 variable

	A	B	C	D
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

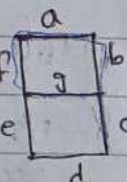
1 - generate eq

a	b	c	d	e	f	g
x	x	x	x	x	x	x
x	x	x	x	x	x	x
x	x	x	x	x	x	x
x	x	x	x	x	x	x

* Truth table (16 bit)

Decimal	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

a	b	c	d	e	f	g
1	1	1	1	1	1	0
0	1	1	0	0	0	0
1	1	0	1	1	0	1
0	1	1	0	0	1	1
1	0	1	1	0	1	1
1	0	1	1	1	1	1
1	1	1	0	0	0	0
1	1	1	1	1	1	1
1	1	1	0	0	1	1
x	x	x	x	x	x	x
x	x	x	x	x	x	x



AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

→ for "A"

$$G_1 \rightarrow A$$

$$G_2 \rightarrow C$$

$$G_3 \rightarrow B'D' \text{ edges}$$

$$G_4 \rightarrow BD$$

$$a = A + C + B'D' + BD$$

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

2) KMAP for "B"

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

$$A + B'D'$$

$$G_1 \rightarrow B'$$

$$G_2 \rightarrow \overline{CD}$$

$$G_3 \rightarrow CD$$

$$b = B' + \overline{CD} + CD$$

3) KMAP for "C"

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	x	x	x	x
10	1	1	x	x

$$G_1 \rightarrow \overline{A}$$

$$G_2 \rightarrow \overline{CD}$$

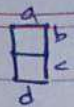
$$C = \overline{A} + \overline{CD} + B + A'D$$

$$c = \overline{C} + B + D$$

4) for "D"

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	x	x	x	x
10	1	1	x	x

$$d = \bar{B}\bar{D} + C\bar{D} + \bar{B}C + B\bar{C}D + A$$



5) for "E"

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

$$e = C\bar{D} + \bar{B}\bar{D}$$

6) for "f"

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$$f = \bar{C}\bar{D} + B\bar{C} + A + B\bar{D}$$

7) for "g"

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$$g = A + B\bar{C} + C\bar{D} + \bar{B}C$$